

Counterparty Credit Risk

- Overview of Modeling Approaches

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Lecture Notes for 2026 Credit Analytics (MATH5450G001)

Agenda

► Overview of Credit Risk

- What is Credit Risk?
- Market Risk, Credit Risk and Counterparty Credit Risk
- History and Lessons (LTCM 1998 and Lehman Brothers 2008)

► Modeling Credit Exposure for Trading Products

- Scenario Generation
- Instrument Valuation
- Portfolio Aggregation

► Market Pricing of Counterparty Credit Risk

- CVA calculation at transaction Level
- Modeling CVA at portfolio level, in presence of netting and/or collateral agreement

► Regulation and Risk Management

- Basel Capital Framework for Counterparty Credit Risk

| 2 – Interagency Guidance on Counterparty Credit Risk Management

What are risks inherent in Trading book?

▶ **Market Risk in Trading Book**

- P&L as result of adverse market moves
- VaR as a measure to monitor and control Market risk

▶ **Credit Risk in Trading Book**

- Issuer Risk, such as in bond held in book
- Settlement Risk, such as in FX spot/forward
- Pre-settlement Risk, such as in FX forward/IR Swap (CCR)

▶ **Counterparty Credit Risk (CCR)**

- It is the risk that one or more parties in a financial transaction will fail to fulfill their side of the contractual agreement, as a result of Counterparty default or its ability to meet the contractual obligations.
- CCR arises primarily from 2 main types of derivatives transactions:
 - OTC derivatives such as interest rate swaps and swaptions.
 - SFT (eg, Securities financial transactions) such as Repo/Reverse Repo where securities are used to borrow cash or vice versa.

Market Risk vs. Credit Risk

Market risk and credit risk are asymmetrical with respect to the underlying market risk factor.

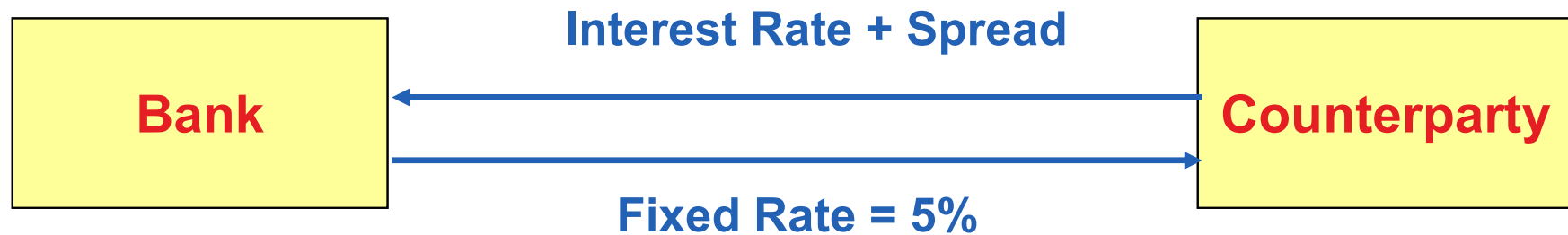
▶ Market Risk

- Risk of losses to the Bank portfolios arising from changes in market prices in Interest Rates, Foreign Exchange rates, Credit spreads, Equity and Commodity prices, etc.

▶ Credit Risk

- Potential loss to the Bank if the Borrower (also known as Counterparty) fails to meet its obligations on the Loans and/or Traded positions (eg, transactions) if our aggregated position with a counterparty has a positive value and the counterparty defaults.

Example: Interest Rate Swap



► Market Risk

- Swap has negative MtM to the Bank, when Interest Rate goes down.
- Bank has **Market risk** on this trade because Bank incurs a loss (eg, negative MtM) when Interest rate moves against the Bank's position in the Swap trade.

► Credit Risk

- Counterparty defaults, Interest Rate goes up and the swap has positive MtM to the Bank.
- Bank has **Credit risk** to Counterparty because Bank could suffer a potential loss of economic value of swap at any future time when Counterparty defaults.

Market and Credit Risks

Attribute	Market Risk	Credit Risk
Time Horizon	▶ Short-Term risk (1-10 days)	▶ Long-Term Risk
Nature of Loss	▶ Adverse Market Move	▶ Loss is contingent on counterparty default when we have trading gains
Risk Mitigation	▶ Matching Positions Risk can be eliminated by offsetting positions	▶ Use of Netting Agreements ▶ Collateral ▶ Mutual Terminations ▶ Counterparty Diversification
Legal Issues	▶ Loss is not dependent on legal issues	▶ Loss magnitude is affected by enforceability of netting agreements



Types of Credit Risks

➤ **Credit Risk = EAD × PD × LGD**

TYPE OF CREDIT RISK

CAUSE OF ECONOMIC LOSS

Lending Risk	▶ Borrower Defaults ▶ Loan value decreases as a result of – Decrease in credit quality of borrower – Increase in general credit spread
Issuer Risk (Specific Risk)	▶ Issuer of security defaults ▶ Security's market value decreases – Decrease in credit quality of borrower
Counterparty Risk • Settlement • Pre-settlement	▶ Counterparty Defaults – In an exchange, we pay but don't receive – Counterparty defaults before settlement and contract has positive value to us

What is Counterparty Risk?

- ▶ Counterparty Risk is credit risk inherent in the derivatives trading

Exhibit 1

Counterparty risk defined

Trading partners, not borrowers, are the source.

Main risk categories		Description	Capital market products		
			Exchange-traded	OTC-traded	Loans
Credit risk	Issuer risk	Risk that issuer/borrower defaults and is not able to fulfill the obligation (eg, unable to make full repayments)	●	● ¹	●
	Counterparty risk	Default risk: risk that counterparty defaults and transaction fails to pay; double-default (or wrong-way) risk occurs when collateral is also impaired		●	
		Replacement risk: after a default, risk that replacing deal under same conditions is not possible		●	
		Settlement risk: risk that party involved in the settlement, such as a correspondent bank, fails before transaction has completely settled		●	
Market risk		Risk that value of investment decreases because of change of market prices	●	●	
Operational risk		Risk of loss resulting from inadequate or failed internal processes, people, and systems or from external events	●	●	●
Liquidity risk		Risk that a given security or asset cannot be traded promptly in the market (eg, to prevent a loss)	●	●	●



Transactions carry Counterparty Risk

► Pre-Settlement Risk

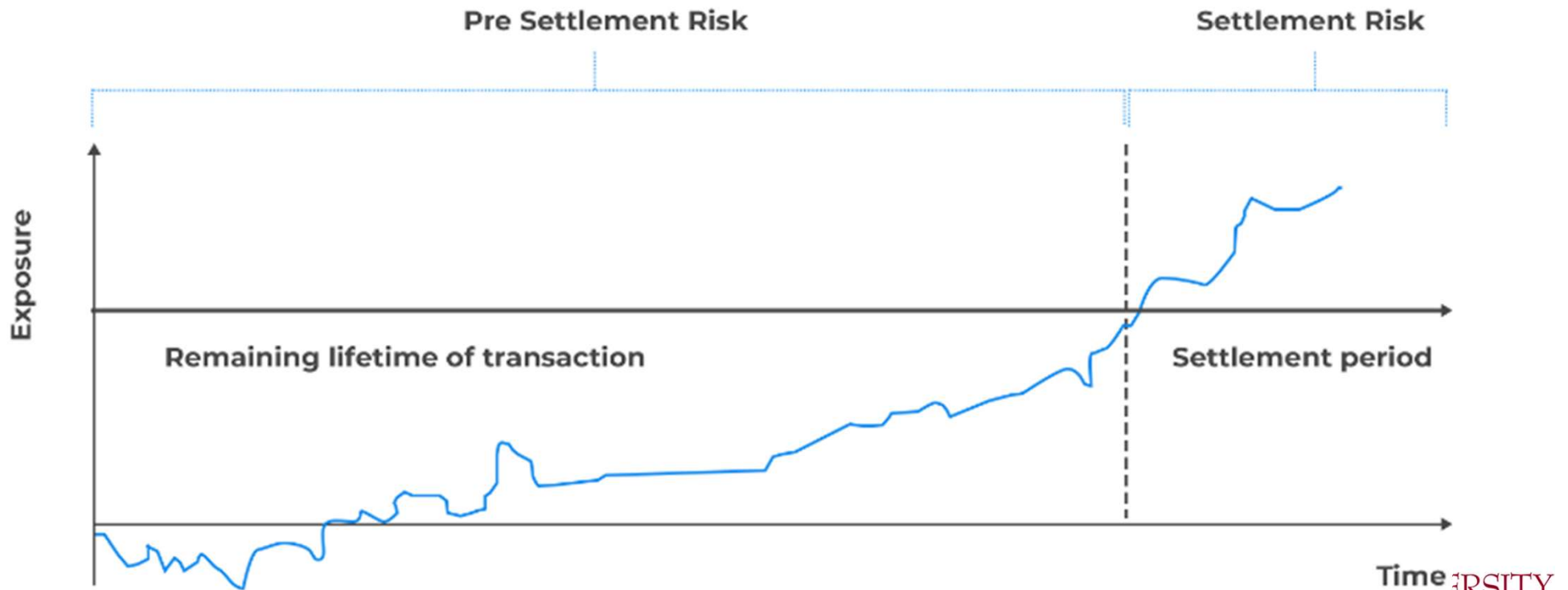
- The risk that a Counterparty will default prior to the final settlement of transaction



► Settlement Risk

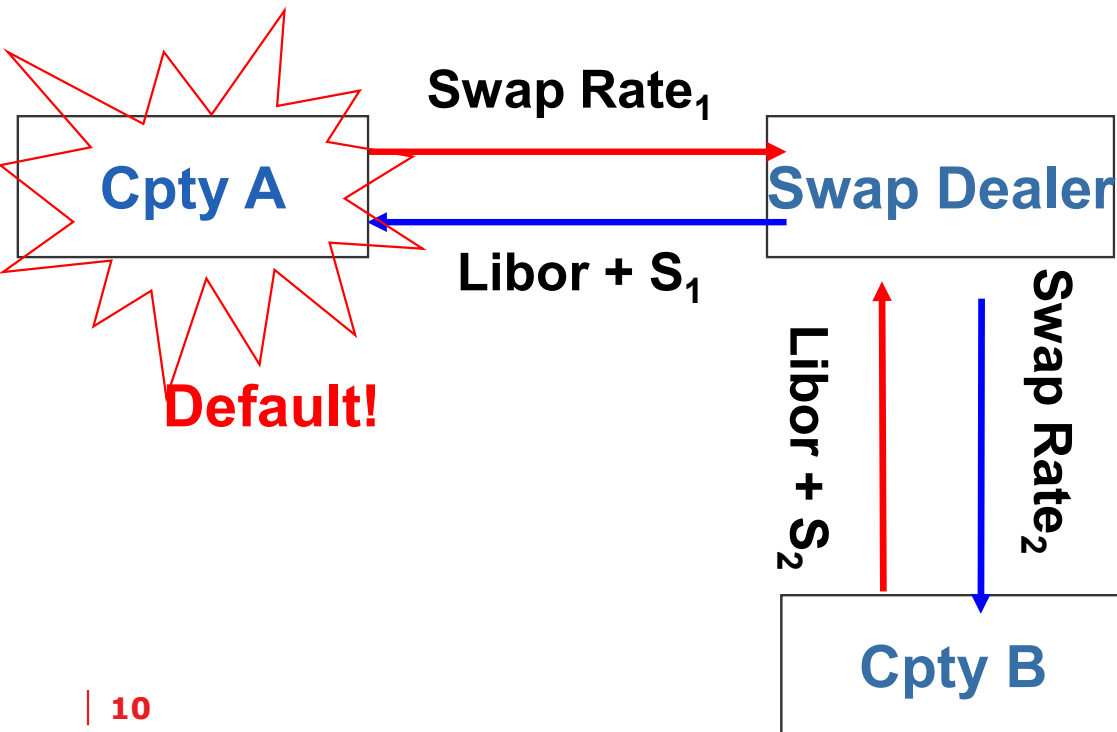
- The risk that a Counterparty fails to settle (pay or deliver) according to the contractual term of transaction

Pre-settlement Risk vs. Settlement Risk

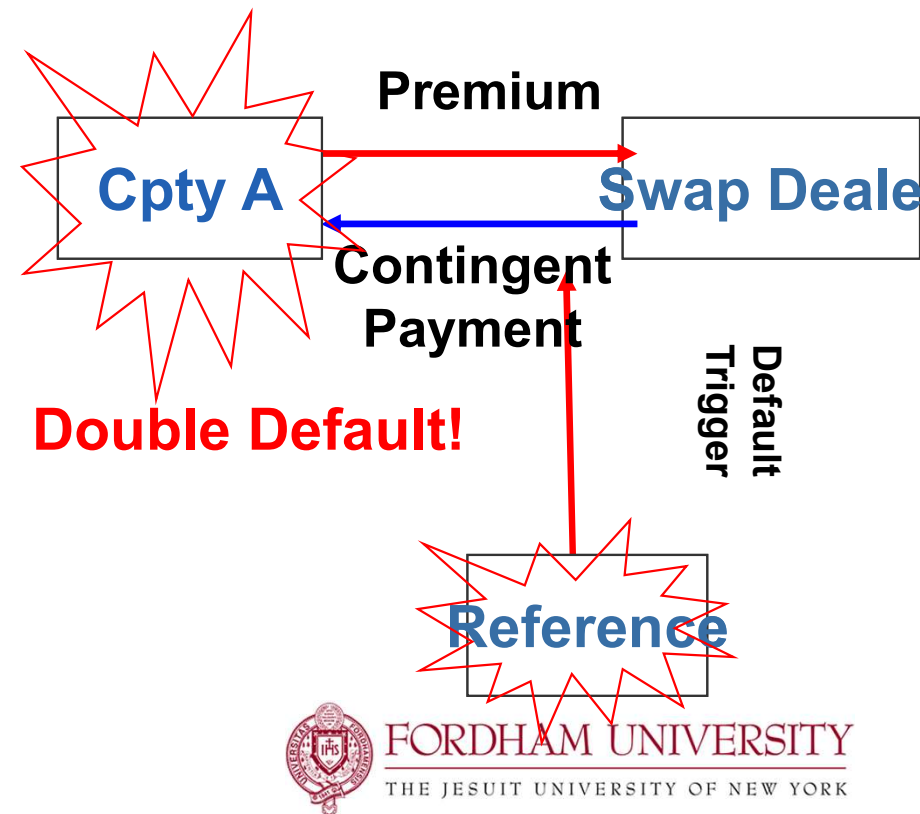


Counterparty Credit Risk - Examples

- ▶ **Interest Rate swaps** carry counterparty risk because Counterparty may fail to make floating/fixed cash payments as and when required. However, risk is reduced because there is no exchange of principal and further reduced through netting/offsetting of the fixed and floating payments exchanged periodically.



- ▶ **Credit Default swaps** carry counterparty risk also in the form of wrong-way risk (WWR) when exposure to Counterparty is adversely correlated with credit quality of that Counterparty.



Counterparty Risk – History and Lessons

- ▶ **What are the Key Lessons in History?**
 - LTCM failure in 1998
 - Failure and Aftermath of Lehman Brothers
 - Bank's exposure to DPC/Monoline guarantees (AIG, MBIA, Ambac) post – 2008 Crisis
 - Recent – Archegos default in March 2021
- ▶ **Chain of Counterparty Failures were observed in 2008**
 - Banks which did not consider CCR, suffered an unexpected large loss
 - Banks which did consider CCR but did not hedge, suffered a large loss, as well
 - Banks which did consider CCR and also hedge, suffered a limited amount of loss
- ▶ **MtM loss due to CVA/CCR were about 2/3rd of total CCR losses with 1/3rd due to actual default during 2008 Financial Crisis**

Counterparty Risk – History and Lessons

Counterparty credit risk management is a critical aspect of financial institutions' risk management practices. Some key lessons learned from recent events

1. **Long-Term Capital Management (LTCM) Crisis (1998):** Twenty-five years ago, the New York Fed facilitated the stabilization of LTCM, a heavily leveraged hedge fund that had borrowed over \$125 billion from its counterparties. The event prompted deep reflection on counterparty credit risk management practices by banks that finance nonbanks' activities. Since then, banks have made progress in improving their practices¹. (Michael Barr on importance of CCR)
2. **Archegos Capital Management (2021):** The failure of Archegos resulted in over \$10 billion of reported losses across several banks. It revealed gaps in how banks manage their exposure to investment funds. Banks have since improved information disclosures, adopted risk-sensitive margin practices, and enhanced risk management tools¹. (Michael Barr on importance of CCR)
3. **Energy Price Jumps (2022):** Unusually large price jumps in energy markets in summer 2022 highlighted weaknesses in banks' counterparty credit risk management. These events underscore the importance of understanding a counterparty's risk profile through advanced risk measurement analytics.
4. **Lehman Brothers (2008):** The collapse of Lehman Brothers exemplified the materialization of counterparty risk. It serves as a historical lesson for risk management³.
5. **FTX (2022):** FTX, a major cryptocurrency exchange, filed for bankruptcy due to a liquidity crisis and allegations of misuse of customer funds. This event also underscores the need for robust counterparty risk management practices³. (OCC)



Counterparty Risk – Key Lesson of 2008

▶ Bear Stearns

- 10bn Collateral sized by JPMorgan Chase

▶ Lehman Brothers

- 15bn Collateral sized by JPMorgan Chase

▶ AIG

- 20bn Collateral shortfall prior to Govt bailout

▶ Monoline Insurance Co

- Bank also suffered large write-down on exposure to DPC/Monoline guarantees

▶ AIG's Main Counterparties

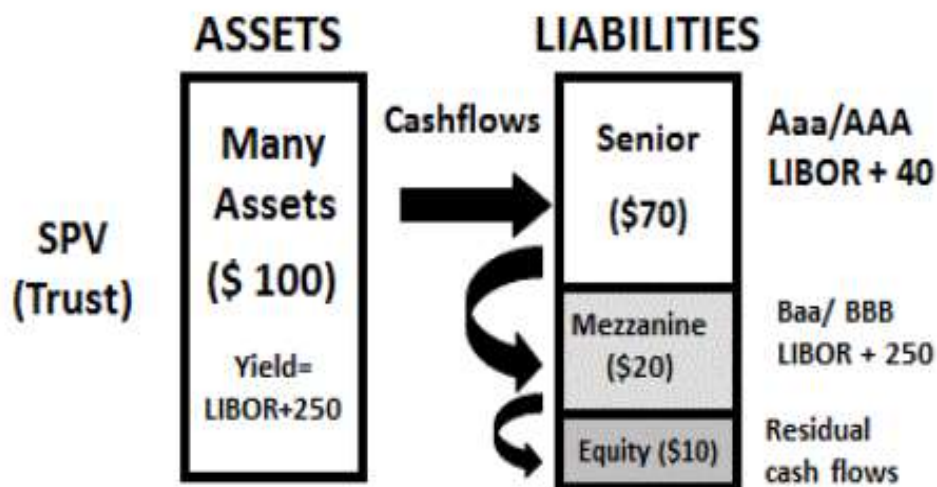
(USD billions)		
Institution	Aggregate Notional value	Negative Mark to Market value
Société Générale	16.4	8.4
Goldman Sachs	14.0	8.0
Deutsche Bank	8.5	3.6
Merrill Lynch	6.2	3.4
Calyon	4.3	2.4
UBS	3.8	2.0
Deutsche Zentral-Genossenschaftsbank (Coral Purchasing)	1.8	1.0
Barclays (BGI Cash Equivalent Fund II and Barclays)	1.5	1.0
Bank of Montreal	1.4	0.6
The Royal Bank of Scotland	1.1	0.6

Morgan Stanley in 2008 Crisis: \$37B Loss

- ▶ **What caused Morgan Stanley's 2008 crisis, in which the bank almost collapsed? Who paid for Morgan Stanley's bailout?**
 - Detailed in "The Big Short" (2010) by Michael Lewis, author of "Liar's Poker" (1989) Rising through the Wreckage on Wall Street

Collateralized Debt Obligations

The Basic Idea



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- ▶ **Mortgage-backed securities (MBS) and Collateralized Debt Obligations (CDO)**

- Howie Hubler, a successful bond trader, made large, complex bets against subprime mortgages using credit default swaps (CDSs).
- To offset the cost of these hedges, he sold insurance on supposedly "safe" AAA-rated collateralized debt obligations (CDOs), which were bundles of these risky mortgages. Trades lost \$9B alone!



LTCM (1998) and Archegos (2021) Collapse

➤ LTCM Crisis (1998)

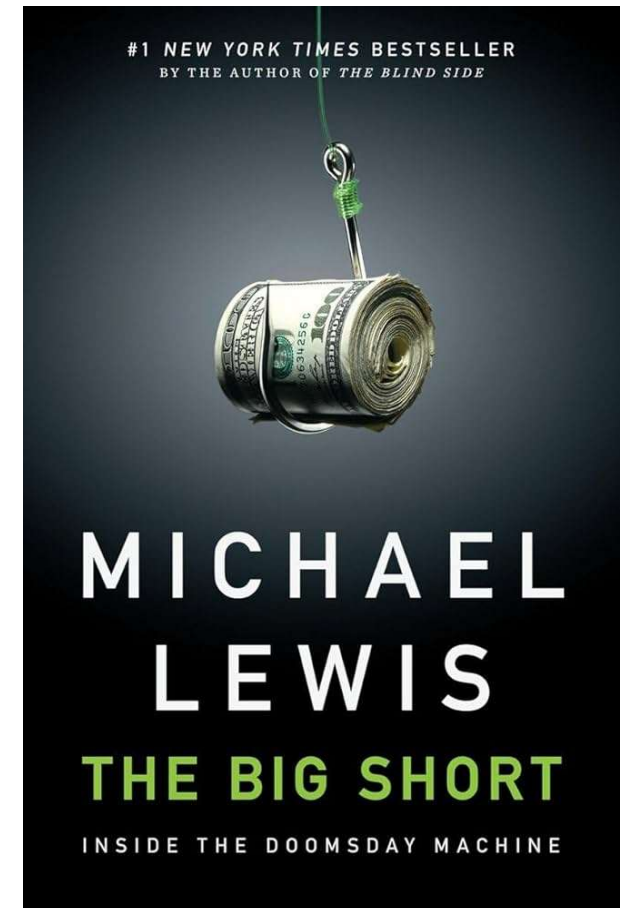
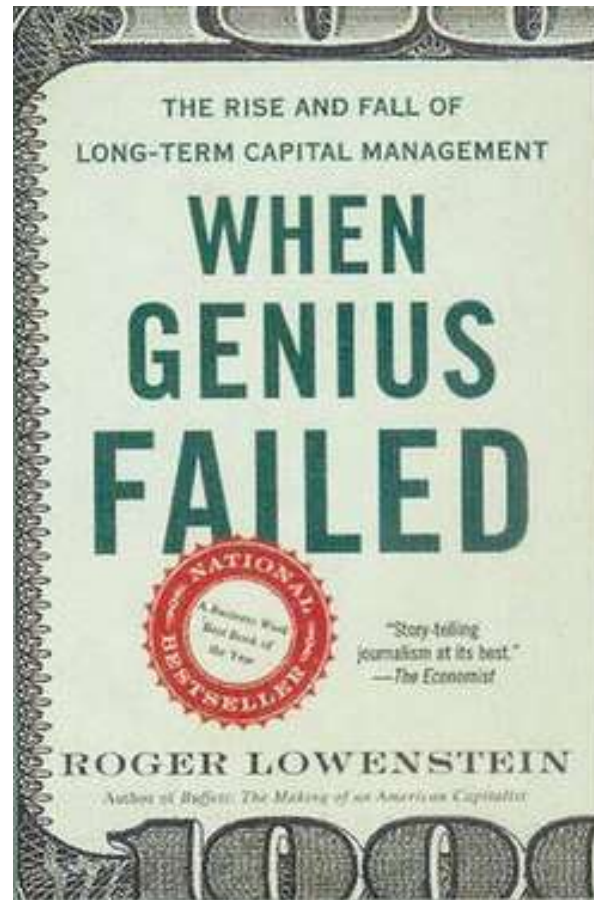
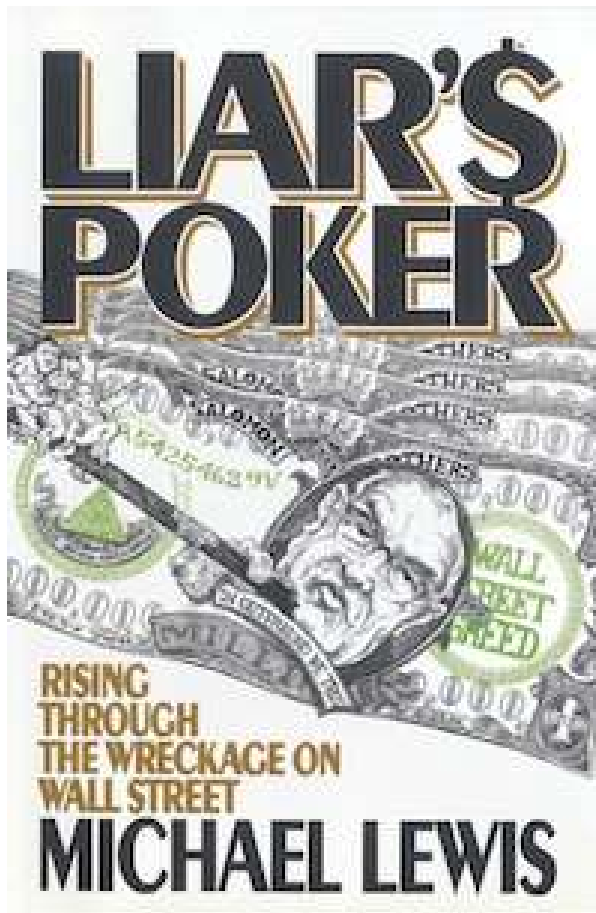
- Lowenstein “*When Genius Failed: Rise and Fall of Long-Term Capital Management*” (2001).
- Model Risk: Value-at-Risk (VaR) model underestimating the risk.
- Liquidity Risk: unexpected correlation or breakdown of historical correlation for long-short convergence trading strategies
- Crisis was averted with bailout fund from a consortium of 14 Wall Street banks organized by NY Fed.

➤ Archegos (2021) and Credit Suisse Collapse

- Archegos had large, concentrated positions on Tech stocks (eg, VaicomCBS) using Total Return Swaps (TRS), which helped to hide exposure from the Lending banks (eg, *Counterparty Risk*)
- Archegos defaulted on margin calls from several global investment banks, including Credit Suisse, Nomura, Morgan Stanley and Goldman Sachs
- Archegos lost \$20B in 2 days when Viacom stock plunged 27%. MS and Nomura reported \$1-2B loss while Credit Suisse lost \$5B leading to forced merger with UBS 3 months later in July 2021.

Recommended Readings

- ▶ #1 Bestsellers in 1980s, 1998 and 2008 crisis



Counterparty Risk Intermediation

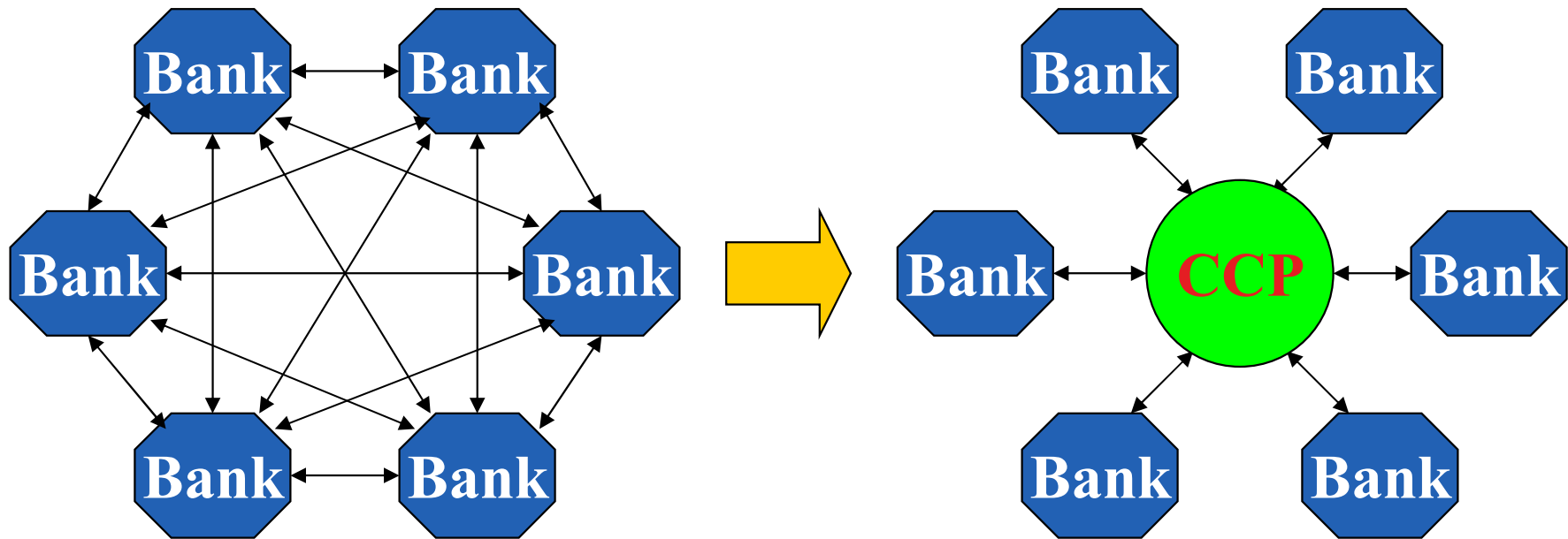
- ▶ Counterparty Risk **intermediation** is a practice of using 3rd-party for risk intermediation and performance guarantee of trading counterparty, to reduce the Counterparty Risk
- ▶ **Special Purpose Vehicles (SPV)**
 - It is designed to establish a bankruptcy-remote entity that gives a counterparty preferential treatment as a creditor in the event of a default. However, SPV introduces legal risk because the envisaged preferential treatment may not materialize following a default event.
- ▶ **Derivative Product Companies (DPC)**
 - Similar to SPV, it is a bankruptcy-remote entity set up to specifically participate in derivative transactions. It differs from an ordinary SPV in that the originating bank injects capital for a strong credit rating (usually triple-A) from credit rating agencies.
- ▶ **Monoline Insurance Company**
 - It is set up as an insurance company to provide the financial guarantees (wraps) for MBS and CDO/CLOs. Compare to DPC, Monoline is not required to post collateral and enjoy slightly more favorable capital requirements and their expected losses are likely to be lower because they generally wrap good quality products.
- ▶ **Central Counterparty (CCP) – OTC Clearing**

Dodd-Frank Reform Act

- Dodd-Frank Financial Reform Act will have a dramatic impact in many aspects of the financial markets:
 - ✓ Migration of significant portions of OTC derivatives trading to Exchanges and collateral and trade settlement through Clearinghouses
 - The regulatory capital impact of this are being determined through the US regulatory process in connection with new Basel III capital framework.
 - ✓ Development of a resolution mechanism to effect orderly liquidation of “systemically important” financial institutions.
 - Note that the determination of whether an institution is “**systemically important**” is left to the discretion of the newly formed Financial Stability Oversight Council.
 - ✓ Elimination of “proprietary trading” by banks under the **Volcker Rule**

OTC Clearing Initiative

- ▶ Standardization of Derivative Contracts
- ▶ Swap Agreements and Counterparties
- ▶ Increased Margin/Collateral Requirement
- ▶ Swap Execution Facilities (SEFs) required to OTC clearing



Derivatives - OTC vs Clearing

- ▶ **What is a derivative? (John Hull: Futures, Options and Derivatives)**
 - A derivative is a financial instrument that derives its value from the performance of an underlying asset (stock and bond) and is used to transfer risk from one party to another.
 - There is a wide range of financial and non-financial instruments that can be used as underlying assets for derivatives, including interest rates, fixed income instruments, foreign currencies, equities and commodities.
- ▶ **As the result of regulatory reforms, Derivatives markets have become safer, more resilient and transparent**
 - **Central clearing:** 91.1% of interest rate derivatives (IRD) traded notional and 83.0% of credit derivatives traded notional in the US was cleared. Market participants have consistently cleared more than what is required under the CFTC's clearing mandate
 - **Margin requirements for non-cleared derivatives:** The amount of regulatory IM has been increasing as margin rules for non-cleared derivatives have been phased-in since September 2016 and more firms and new transactions have become subject to the requirements. The most recent ISDA Margin Survey found that the 20 largest global market participants collected approximately \$173.2 billion of IM and \$897.3 billion of VM for their non-cleared derivatives transactions



Derivatives - Use and Misuse

- ▶ **Example 1 - Apple** uses derivatives to partially offset its business exposure to foreign currency and interest rate risk on expected future cashflows, net investments in certain foreign subsidiaries, for assets and liabilities management (ALM).
 - The company enters into interest rate swaps (IRS) to manage interest rate risk on its outstanding term debt.
 - The company also uses currency derivatives such as forwards and cross-currency swaps to manage and protect its investment from fluctuations in foreign currency exchange rates
- ▶ **Example 2 - Chevron** is exposed to market risks related to the price volatility of crude oil, refined products, natural gas, natural gas liquids, liquefied natural gas and refinery feedstocks.
 - It uses commodity derivatives to manage these exposures on a portion of its activity, including firm commitments and anticipated transactions for the purchase, sale and storage of crude oil, refined products, natural gas, natural gas liquids and feedstock for company refineries.
 - The company also uses commodity derivatives for limited trading purposes¹

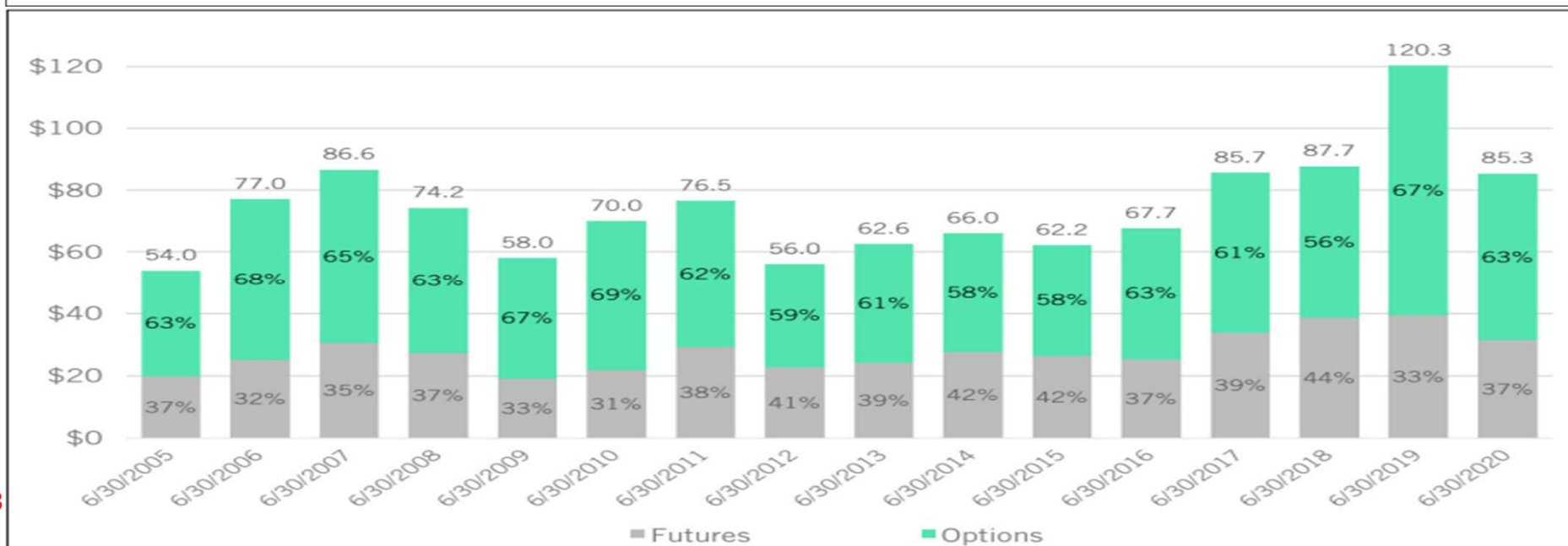
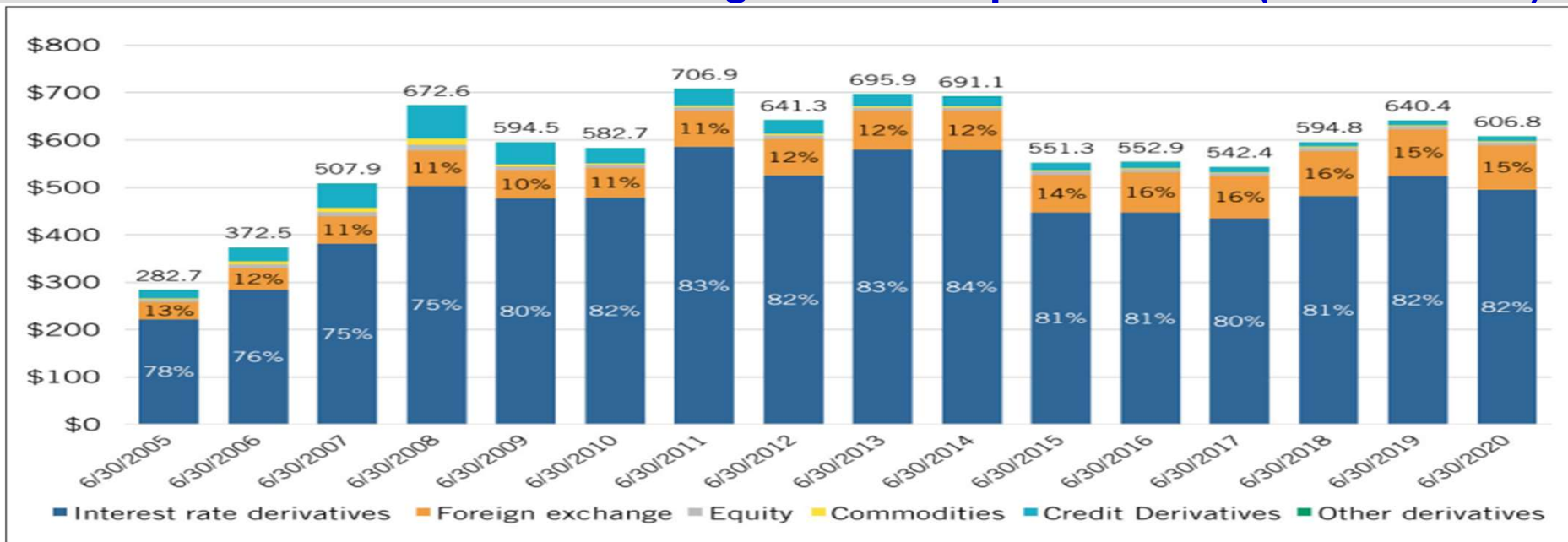
Derivative Contracts - Overview

► Examples of Derivative Contract Types

UNDERLYING	CONTRACT TYPES				
	Exchange-traded futures	Exchange-traded options	OTC swap	OTC forward	OTC option
Equity	DJIA Index future Single-stock future	Option on DJIA Index future Single-share option	Equity swap	Back-to-back Repurchase agreement	Stock option Warrant Turbo warrant
Interest rate	Eurodollar future Euribor future	Option on Eurodollar future Option on Euribor future	Interest rate swap	Forward rate agreement	Interest rate cap and floor Swaption Basis swap Bond option
Credit	Bond future	Option on Bond future	Credit default swap Total return swap	Repurchase agreement	Credit default option
Foreign exchange	Currency future	Option on currency future	Currency swap	Currency forward	Currency option
Commodity	WTI crude oil futures	Weather derivative	Commodity swap	Iron ore forward contract	Gold option

Derivative Contracts - Overview

Global OTC Notional Outstanding and ETD Open Interest (US\$ trillions)



Derivative Contracts - Overview

► Interest Rate Derivatives Traded vs Cleared as % Notional (US\$ trillions)



Derivative Contracts - Overview

► Credit Derivatives Traded vs Cleared as % Notional (US\$ trillions)

